

$$\mathbb{Z} \cdot w = w$$

$$E_{\mathbb{Z} \cdot w} = \{ \underset{\uparrow 1}{u_1}, \underset{\uparrow 2}{u_2}, \underset{\uparrow 3}{u_3}, \underset{\uparrow 4}{u_4}, \underset{\uparrow 5}{u_5}, \underset{\uparrow 6}{u_6}, \dots \}$$

Considérons



$$f: \mathbb{Z} \cdot w \rightarrow \mathbb{N}$$

$$u_i \mapsto |f(u_i)| = \begin{cases} 2i-1 & \text{si } i \geq 1 \\ 2i & \text{si } i \geq 2 \end{cases}$$

Il est clair que f est un isomorphisme de $\mathbb{Z} \cdot w$ vers $\mathbb{N}$

$$E_{\mathbb{Z} \cdot w} = \{ 0, 1, \dots, w, w+1, \dots \} \\ = \mathbb{N} + \{ w+n, n \in \mathbb{N} \}$$

$$E_{\mathbb{Z} \cdot w} = \{ \underbrace{0, 1, \dots}_{w}, \underbrace{w, w+1, \dots}_{w+1}, \underbrace{w+2, w+3, \dots}_{w+2}, \dots \}$$

$$= \mathbb{N} + \{ w+n, n \in \mathbb{N} \} + \{ w+2n, n \in \mathbb{N} \} + \dots$$

$$E_{w \cdot w} = IN + \sum_{i=1}^{\infty} \{w \cdot i + n, n \in \mathbb{N}\}$$

$$\begin{aligned} E_{w^d \cdot n} &= E_{w^d} + \{w^d + n, n \in \mathbb{N}\} \quad n \geq 2 \\ &\quad + \{w^d \cdot 2 + n, n \in \mathbb{N}\} \\ &\quad \vdots \\ &\quad + \{w^d \cdot (m-1) + n, n \in \mathbb{N}\} \end{aligned}$$

$$\begin{aligned} &= \{0, 1, \dots, w, w+1, \dots, w^2, w^2+1, \dots, \\ &\quad w^d, w^d+1, \dots, w^{d-2}, w^{d-2}+1, \dots, \\ &\quad w^{d-(m-1)}, w^{d-(m-1)}+1, \dots\} \end{aligned}$$

$$E_{w^d} = E_{w^{d-1}} + \sum_{i=1}^{\infty} \{w^{d-1} \cdot i + n, n \in \mathbb{N}\}$$

$$\begin{aligned} E_{w^d} &= E_{w^{d-1}} + \sum_{i=1}^{\infty} \{w^{d-1} \cdot i + n, n \in \mathbb{N}\} \\ &= \{0, 1, \dots, w^{d-1}, w^{d-1}+1, \dots, \\ &\quad w^{d-2}, w^{d-2}+1, \dots, w^d, w^d+1, \dots\} \end{aligned}$$